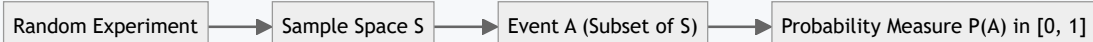


Day 20 - Lecture 20.2: Core Foundations of Probability

[Python 3.10+](#)[Jupyter Notebook](#)[PDF notes.pdf](#)[Language: English | हिंदी / Hinglish | !\[\]\(e3f8612927870f2e0f9f5989e6dd3064_img.jpg\) Back to Lecture 20 Module](#)

1. Random Experiments, Sample Space, and Events

A **random experiment** is an observational process whose outcome cannot be predicted with absolute certainty prior to execution, but whose set of all possible outcomes is known in advance.



Core Definitions:

- Sample Space ($\mathcal{S}$ or Ω):** The set containing all mutually exclusive, exhaustive primitive outcomes ω of a random experiment.
 - Example (Fair 6-sided die): $\mathcal{S} = 1, 2, 3, 4, 5, 6$.
 - Example (Two successive coin tosses): $\mathcal{S} = HH, HT, TH, TT$.
- Event ($\mathcal{E}$ or A):** Any subset of the sample space ($A \subseteq \mathcal{S}$).
 - Example: An even roll on a die: $A = 2, 4, 6 \subseteq \mathcal{S}$.
- Null Event ($\emptyset$):** The empty set representing an impossible outcome ($P(\emptyset) = 0$).
- Certain Event ($\mathcal{S}$):** The entire sample space representing an inevitable outcome ($P(\mathcal{S}) = 1$).

2. Kolmogorov's Axioms of Probability

Modern probability theory rests upon three fundamental axioms established by Andrey Kolmogorov in 1933. Let $\mathcal{S}$ be the sample space and $\mathcal{F}$ be the event space (σ -algebra of subsets):

Axiom 1: Non-Negativity

For every event $A \subseteq \mathcal{S}$, the assigned probability is non-negative:

$$\boxed{P(A) \geq 0}$$

Axiom 2: Unit Measure (Normalization)

The probability of the entire sample space is equal to unity:

$$\boxed{P(\mathcal{S}) = 1.0}$$

Axiom 3: Countable Additivity (Disjoint Events)

If $A_1, A_2, A_3, \dots$ is a countable sequence of pairwise mutually exclusive (disjoint) events such that $A_i \cap A_j = \emptyset$ for all $i \neq j$, then:

$$\boxed{P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)}$$

3. Fundamental Derivative Properties

From Kolmogorov's axioms, standard probability calculus theorems are derived:

$$\boxed{P(\emptyset) = 0.0}$$

$$\boxed{0 \leq P(A) \leq 1.0 \quad \forall A \subseteq \mathcal{S}}$$

$$\boxed{A \subseteq B \implies P(A) \leq P(B) \quad \text{ext}\{(\text{Monotonicity})\}}$$

4. Python Implementation: Verifying Kolmogorov's Axioms

```
import numpy as np

# Sample space for a standard 6-sided fair die
sample_space = {1, 2, 3, 4, 5, 6}
n_s = len(sample_space)

# Define events
event_even = {2, 4, 6}
event_prime = {2, 3, 5}
event_odd = {1, 3, 5}

# Probability measure for uniform discrete sample space
def prob(event, s=sample_space):
    return len(event.intersection(s)) / len(s)

# Axiom 1: Non-negativity
print(f"P(Event) = {prob(event_even)} >= 0: {prob(event_even) >= 0}")

# Axiom 2: Normalization
print(f"P(S) = {prob(sample_space)} == 1.0: {prob(sample_space) == 1.0}")

# Axiom 3: Additivity of disjoint events (Even and Odd are disjoint)
is_disjoint = len(event_even.intersection(event_odd)) == 0
union_event = event_even.union(event_odd)
p_union = prob(union_event)
p_sum = prob(event_even) + prob(event_odd)

print(f"Events Disjoint: {is_disjoint}")
print(f"P(Event U Odd) = {p_union:.4f}, P(Event) + P(Odd) = {p_sum:.4f}")
print(f"Additivity holds: {np.isclose(p_union, p_sum)}")
```

5. Key Takeaways & Interview Points

- **Probability is a Measure:** Probability maps subsets of sample space to the closed interval $[0, 1]$.
- **Kolmogorov Foundation:** All advanced machine learning probability theorems (Bayes, CLT, Markov chains) are derived directly from Kolmogorov's three axioms.

- **Event vs Outcome:** An outcome is an elementary point $\omega \in \mathcal{S}$; an event is a set of outcomes $A \subseteq \mathcal{S}$.